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CSC 317 — Past Questions & Materials
by Renance

Course: CSC 317

Institution: Federal University of Petroleum Resources, Effurun

Papers in this pack: 2

Levels: 300

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Paper 1 of 2 — CSC 317 Research Methodology 2025/2026

Level 300 • First Semester • PastQuestion

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 Department of Computer Science
 1st Semester: BSc (Computer Science Examination) 2025/2026 Session
CSC317 - RESEARCH METHODOLOGY
 Instruction: Attempt Questions 1, 2 AND ANY OTHER 2
 Time Allowed: 3Hrs

Question 1:

- Describe the aim of a Theoretical/Conceptual Review 5mks
- Describe the role of the Review of Related Literature using the concept of the meta-analysis table often captioned Table 2.1 5mks
- With knowledge gaps identified, how do we resolve the challenges identified and stated using the summary of Related Literature captioned Table 2.2. 5mks
- Describe the 8-steps in conducting a Research 8mks
- Draw a sample framework for a timeline of your Project 7mks

Question 2:

- What should a good background of study consist? 2mks
- Describe the term S.M.A.R.T features of a study's AIM / Objective(s) 5mks
- What makes up the basic elements for the specific objectives of study 5mks
- Describe Hypothesis and Research Question from a researcher's viewpoint 4mks
- Describe what a reader expects out of an Abstract 4mks

Question 3:

- Describe the form for the APA referencing style with an example 2mks
- Describe the form for the IEEE referencing style with an example 2mks
- Describe the form for the Harvard referencing style with an example 2mks
- Describe the form for the Vancouver referencing style with an example 2mks
- Describe the form for the MLA referencing style with an example 2mks

Question 4:

- Describe the concept of Research 2mks
- State the functions of a Research 2mks
- State any 2 Characteristics of Research 2mks
- Draw the knowledge Discovery in Research Data 2mks
- Use a sample ToDo List workplan to meeting your supervisor/mentor 2mks

Question 5:

- What makes up the contents of the Description of an existing system 2mks
- With the existing system in computing science - how can we effectively describe the Proposed system 2mks
- State clearly and concisely what makes up the Data Gathering section 2mks
- Draw the research process 4mks

Question 6:
Describe the following:

- System Requirements / Experimental Testbeds 2mks
- Discussion of Findings 2mks
- Performance Evaluation 2mks
- Benchmarking / Ablation Studies 2mks
- System Conversion/Changeover Procedures 2mks

OJUGO, Arnold Adimabua Ph.D - Examiner

Paper 2 of 2 — CSC 317 Materials

Level 300 • First Semester • Handout

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Outcome-based Research and Reporting

Lecture 02 – Formulating a Problem

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We noted in our previous lecture: a critical step in research is formulation of a research problem. You can effectively do this through and be on your way to better research by accomplishing 4-tasks:

(a) review of literature, (b) formulate the problem, (c) identify the variables, and (d) constructing hypotheses.

Review of Literature or Literature Review (LR)

Existing literature gets us acquainted with available body of knowledge in the area of interest.

Undertaking literature review can be quite pains-taking, time-consuming and sometimes – daunting and frustrating. However, to successfully perform research, we must engage in literature review.

Literature Review helps us accomplish three things:

□ Brings about Clarity: LR is a paradox and a conundrum. Effective LR requires you must have an idea of the problem you wish to investigate; And, literature review also plays a critical role in shaping your research because it helps you to also better understand the subject area so that you can better conceptualize your problem clearly, concisely and precisely. Thus, it brings you to better knowledge of your research problem and the available body of knowledge in the area.

□ Improves your Methodology: LR tells you if other authors have used procedures and methods similar to the one you are proposing, which procedures and methods worked well, and the problems they faced. Thus, aware of these – you are better positioned to select a methodology that is capable of providing answers to your research problem. This will increase your confidence in the methodology you are proposing and equip you well enough to defend it also.

□ Broadens your Knowledge: Literature review ensures that you read around the subject area. It is extremely important you know what other researchers have done, what they found in regards to similar questions and problems, what theories have been put forward/used, and what gaps currently exists in the relevant body of knowledge (BOK). A thorough literature review

helps to ensure that you fulfill this expectations. It also helps you to understand how the findings of your study fits into the existing body of knowledge (BOK)

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Procedures for Review of Literatures

Your review of literature in a broad area of interest should aim to gradually help you narrow to what you wish to find out about. The danger of not having a specific idea of what you wish to do – is that it may condition your mind and thinking to less innovative choice of the research problem and methodology. Please, try to conceptualize your problem before undertaking LR. LR is a continuous process that often begins before a research problem is formulated and continues till the report is finished. This is because you will continue to do so as you integrate the findings from your study with those of others, stating whether your findings support or contradicts other studies. This comparison is also an integral part of your report.

There are usually 4-steps in conducting LR

□ Step 1: Search for BOK: You must first, have an idea of the broad subject area, and the problem you wish to investigate. They will help you set the parameters for your search. As you select (download) the materials to use – you must compile a bibliography. There are 2-major sources to use namely: books and (peer-reviewed) journal articles. Please, do not use Internet sources, magazines, newspapers, wikipedia, blogs etc – as a source for your work. Compiling a bibliography can be easily done using Mendeley, EndNote, Zotero, X9 etc.

□ Step 2: Review the Literature: Read the BOK & extract theme/issue that belong together.

Develop a theoretical framework, using separate sheets for each book/journal. As you read further, slot each information where it logically belongs under the headings so far developed.

Add more headings where possible and critically read each reference to note the following:

- a. Is the knowledge relevant to my theoretical framework and affirms beyond doubts
- b. Note the theories, criticisms and their biases – for the adopted methodologies (i.e. study design, sample size and characteristics, measurement procedures etc) and the criticisms of each
- c. Examine to what extent the findings can be generalized to other situations
- d. Notice where there are significant differences of opinion among researchers and give your opinion about the validity of these differences
- e. Ascertain areas where little or nothing is known – the gaps that exists in the BOK.

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Procedures for Review of Literatures...Continued

□ Step 3: Develop a Framework: LR is a continuous task. But, you only have limited time. So, you must set parameters to review literature in relation to themes pertinent to your topic. As we read, you discover that the problem you wish to investigate has its roots in a number of theories

that are developed from different views. You must then sort the information from the books and journals you need under main themes (theories) that highlights dis(agreements) among authors and identifies gaps or unanswered questions. Some literature will deal in(directly) with your topic. Use these as a basis for developing your theoretical framework – to guide you as you read. So, you must read some literature, and then attempt to develop a framework (even when it is weak) within which you will organize the rest of your LR. As you read more, you may likely change your framework. For example, if you wish to study the relationship between personality traits and media presence as factors affecting the susceptibility of users to phishing attacks

a. Traits: theories, trends, critique indices, factors affecting traits, methods of controlling traits, reasons for the

acceptance of traits, etc

b. User Media presence – theories, trends, indices for critiques, why users stay online longer, social media

presence, factors to slow down on media presence, etc

c. User susceptibility – theories, trends, indices for critiques, factors affecting user susceptibility, etc

d. Phishing – theories, trends, methods of phishing that explains the relationship between media presence and

personality traits, and how these impact on phishing, and implication of this relationship.

□ Step 4: Develop Conceptual framework: Every conceptual framework stems from the theoretical framework and concentrates usually on only one section of theoretical framework. A theoretical framework consists of theories, issues, implications, relations etc in which your study is embedded; while, the conceptual framework describes the selected aspects from your theoretical framework that relates specifically to the specific research problem.

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Writing up the Literature Reviewed

Please, refrain from writing the LR under a single heading. Do well to write the LR under and around themes that have emerged from your reading the literature. The headings displaying the themes should be precise, descriptive of the contents, and should follow a logical progression. Findings from the LR should be organized under these themes, providing references through in-text citations to substantiate facts that have been found while you were reading, or to contradict. Your arguments should be conceptually clear, highlighting the reasons for and against the topic area, and referring to the main findings, gaps and issues.

An example can be seen as thus:

Chapter Two: Literature Review

2.1. Theoretical Review

2.2. Conceptual Review

2.3. Review of Related Literature (based on the conceptual review) generated as Table 2.1 on next page

2.4. Gaps Identified and Techniques to Resolving Them

2.5. Summary of Literature Review (generated as Table 2.2. on next page)

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2.2. Review of Related Literature Review – sample table 2.1. The researcher must have researched as in table between 15-to-25 previous authored works and show them in table.

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Prepared by A.A. Ojugo, Ph.D

2.3. Summary of Related Literature

Researcher should identify the components that informed the success of previous works. These features as components should first be described to identify reasons they were used. Research should list between 10-to-15 selected previous authored works as in table.

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Formulating the Research Problem

Not all questions can be transformed into research problem and some may prove intractable.

Formulating a problem requires a considerable knowledge of both the subject area and research methodology. To formulate a research problem therefore – you must: (a) identify the problem (i.e. identifying your destination – for without it, we cannot identify the shortest route to a clear and economical plan for the problem), and (b) specify the problem so that it can withstand scientific scrutiny in terms of procedures required to be undertaken (i.e. a well-formulated problem will enable

us follow properly). You must have a clear idea of what it is you want to find out, and not what you think you must find. A research problem can take various form from simple to very complex; But, the

way you formulate a problem determines almost every step that follows: the study design type to be

used, type of sampling, , research instrument, and type of analysis. Formulation is like input; while, the output – which is the quality of the contents of the research report and the validity of the associations or causations established – is solely dependent on the input. Remember, confusion is but the first towards clarity. So, when you are confused, relax and breathe.

Sources of Research Problems

Many disciplines (social science and humanities) are centered around 4Ps (i.e. people, problems, phenomenon and programs).The general practice are based on atleast, combining 2 or more Ps. They select a group of individuals, to examine the existence of certain issues or to establish the

prevalence of a phenomenon etc. But, research in Computer Science centers around: creativity (artificial life), data (big-data and data science), abstractions, algorithms, programming, the Internet,

and global impact. The emphasis, whatever the direction is explore, discover and establish association or cause-and-effects of some independent variable as they act upon a dependent variable.

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Considerations for Selecting a Research Problem

Some considerations when selecting a research problem that can help you ensure that your study remains manageable and keep you focused/motivated includes:

- a. Interest select a topic that interests you as research can be time-consuming, tedious and often, involves unforeseen problems too difficult to sustain your motivation.
- b. Magnitude you should have sufficient knowledge about research process to visualize the work before you begin. A proposal can help you accomplish this to know before hand what you wish to go up against.
- c. Measurement of concepts – if you are using a concept – be sure you are clear about its indicators and their measurements. Do not use concepts you cannot or unsure can be measured. You can develop a measurement procedure as the study progresses.
- d. Expertise – make sure to have adequate knowledge for the task you are proposing. Allow for the fact that you will learn and may receive help from your mentor and others. But, you must do most of the work.
- e. Relevance – select a topic relevant to your profession. Ensure that the study adds to an existing BOK, bridges current gaps or is useful in policy formulation.
- f. Data Availability – if your topic entails collection of data from secondary sources – before finalizing the topic, make sure you can retrieve the data and they are in the format you want.
- g. Ethical Issues – in formulating a research problem, consider the ethical issues involved. In the course of your study, the sample population may be adversely affected by some questions; deprived of an intervention, expected to share sensitive and private data, or expected to be experimental guinea-pigs. You must examine how ethical issues impacts a sample population as well as known how to resolve such issue where such to arise. You must examine these thoroughly and be sure to grant confidence of non-disclosure to those whom you wish to retrieve the data – at the problem formation stage.

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Steps in Formulation of Research Problem

In choosing a research problem, 2-crucial determinants to keep in mind are: (a) your interest in the

subject area, and (b) manageability of the study within your own constraints. However, the following

steps are to serve as a guide in formulating a research problem:

- a. Identify a broad area of interest in your academic field.
- b. Dissect the broad area into sub-area (you can do so through a brainstorming session with yourself, your peers, fellow professionals and others).
- c. Select a sub-area or areas in which you will like to conduct your research. Start with a process of elimination.
- d. Raise the Research Question that you will like to answer through your study.
- e. Formulate Objectives through main themes, sub-themes etc – for your study (see the section on formulation of objectives on next page).
- f. Assess these objectives to ascertain the feasibility of attaining them in light of the time available to you, the available resources (i.e. finance and human experts), and the technical expertise you have in the field to be researched. Always assess these options that are at your disposal and make informed decisions therein.
- g. Double-Check that you are sufficiently interested in the study and have adequate data and resources to undertake this research. Ask yourself: am I really enthusiastic about this study? Do I really have enough resources for it? If and when your answer to these questions becomes NO – re-assess your objectives and re-start formulation of the problem and your objectives again.

Most researches have 2-aspects: (a) a research problem (i.e. basis for the inquiry), and (b) a study population (i.e. source of information). As you narrow down the research problem – you must extremely identify the source of your information in order to select the appropriate respondents vis-à-vis participants for the study.

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Formulation of Objectives

Objectives are the goals you set out to attain in your study – and they tell the reader what you want to achieve through the study. For this reason, your objectives must show clarity in its being Specific,

Measureable, Achievable, Robust and Time-Dependent (S.M.A.R.T). Objectives should be either:

- a. Main Objectives: Also called the aim – it is the overall statement that portrays the thrust, cardinal points or focus of the study. It creates the association and relation for milestones and deliverables that you seek to discover and establish in your study.
- b. Sub-objectives are specific deliverables and milestones that you wish to investigate within the main framework of your study. So, sub-objectives should: (1) be numerically listed, (2) be clear worded without ambiguities, (3) each sub-objective must contain only one aspect of the study, (4) should start with action oriented words such as ‘to find out’, ‘to determine’, ‘to ascertain’, ‘to measure’, ‘to explore’ etc

How a main/sub-objective is worded determines how a research is classified (i.e. descriptive,

correlational, or experimental). The wordings determines the research design you will adopt. But, irrespective of the research type – objectives must be clear, complete and specifically communicate to a reader your intentions. No ambiguities, non-specificity or incompleteness, either in the wording of your objectives or in the ideas they communicate as in figure below.

If your study is descriptive – your main objective must clearly describe the focus of your study, mention the organization and its location (unless they are to be kept confidential). If correlational, in addition to [describe the focus] + [mention an organization] + [location] – wordings should include

the main variables being correlated (e.g to ascertain impact of migration on social distancing with covid-19 propagation, or to compare effectiveness of different teaching methods on comprehension of students). If experimental, in addition to [describe focus] + [mention an organization] + [location]

+ [correlation variables] – it should indicate the direction of the relationship being tested. E.g. demonstrate provision of maternal and child services to rural Nigeria will reduce infant mortality.

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Figure 1. Characteristics of Research Objectives

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Figure 2. Formulating a Research Problem – the effects of alcoholism

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Figure 2b. Formulating a Research Problem: This example shows the relationship between Fertility and Mortality

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Figure 3. Narrowing a Research Problem

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Variables and Types of Variables

A variable is a feature that can take values and has potential to change (i.e. it can reflect a relations or outcome). There are various types of variables used in research study. These include:

- a. Dependent variable is a variable being measure in an experiment. Thus, researchers do not directly control this variable. Instead, they hope to learn something about the relationship between the different variables by observing how the dependent variable reacts under different circumstances. Also called response/outcome variable, its purpose is to show a response to change in other variables, displaying the outcome. E.g. as in our previous examples is phishing attacks. Many features can influence phishing attacks such as media presence, personality traits, demographics etc.
- b. Independent variable are those the researcher controls and manipulates within the study. In order to determine the cause-and-effect association of a dependent in relation to the independent variable – the researcher must willfully change and alter an independent variable in other to observe when-why-what-and-how the dependent variables changes in response to such alterations. Also called explanatory/predictor variables, they help a researcher to understand their true relationship with dependent variables.
- c. Confounding variables can change and alter the nature of a dependent variable despite not being an independent variable being studied.

Researchers are always seeking to manipulate and/or measure the impact of independent variables on the dependent variable.

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Questions

1. Describe the 4-tasks that must be accomplished to formulate a research problem
2. Describe the 4-steps in conducting a LR and state 3-thing that LR helps you achieve
3. Give a sample framework of a proposed Chapter 2 using any topic of choice
4. List any 7-sources of research problem in Computing Sciences that you know
5. Describe the types of variables used in a research problem
6. When selecting a research problem, list the factors to be considered
7. Describe in your own words, the steps in formulation of a research problem
8. Distinguish between main objective (aim) and sub-objectives (objectives)
9. Identify wordings used for the research types as follows: (a) descriptive, (b) correlational, (c) experimental

Practical Session

To effectively put to use, what we have learned so far – we must accomplish the following:

1. Select a broad area of interest and work out a topic that we wish to investigate on
2. Download 30-45 materials in PDF format (for Journal Articles and Books)
3. Load them into the Mendeley Desktop Reference Manager

4. Create Table 2.1 and Tables 2.2 respectively.
5. Formulate a Research Problem
6. Formulate a Research Objectives
7. State what kind of Research Design to use for your study and rationalize it.

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Installing / Using Mendeley Reference Manager

To effectively manage in-text citations, references and bibliographies used in your research, you can use the Mendeley Reference Manager

1. Download and Install the Mendeley Desktop as in Figure 1 pointing to Figure 2
2. Register for Elsevier via (your user email and chosen password which must consists of uppercase and lowercase characters, digits and special symbol). E.g Betsy1999* as in Figure 2
3. Log-in the created Mendeley account
4. Create a folder for the Project (where you will load all the reference materials on to)
5. Download the MS-Word Mendeley o-cite plug-in
6. Open your MS-Office Word and Go to the Reference Tab
7. Open the Folder where you have downloaded the PDF materials (books and journals)
8. Drag and drop from the folder on to the opened Mendeley Desktop Application

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Outcome-based Research
and Reporting

Lecture 01 – Introduction to Research and
Research Processes

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Part 01 – Outcome-Based Research / Research Methodology

Background: Research has now become an integral part in our daily lives. It is found in the way we ask questions, which we can develop into skillsets required to achieve our tasks daily. It is a way of thinking. It is a quest to resolve and surmount our daily tasks (challenges and risks) so as to yield a competent result that withstands time. Many of our vague idea about research – is often said to involve experimentation, computations and presentations of result to dazzle. The word research is composed of two syllabus. Dictionary defines re as a prefix meaning “again, anew or over-again”, and search is the verb which means to examine closely and carefully, to test and try, or to probe and inquire. Together – they form a noun describing the careful, systematic, patient study and investigation in some field of knowledge, undertaken to establish principles or facts. It is also a structured inquiry that utilizes acceptable scientific methodology to solve problems and create new knowledge that is generally acceptable.

Scientific methods consists of observations, classifications, and interpretations of data. And since research is engaged by almost everyone daily as a social process – the main difference between our day-to-day generalizations and the conclusions usually recognized as scientific methods lies in the degree of formality, rigorousness, verifiability, and general validation of the scientific means and

processes engaged during the arrival at these conclusions. Thus, a research can simply be a systematic investigation to find answers to a problem. A scientific research is a systematic, controlled empirical and critical investigation of propositions about the presumed relations about various phenomena; while, sociological research is primarily committed to establishing systematic, reliable and valid knowledge about the social world.

Research is an out-of-the-box, critical, rationale thinking that provide us simple solutions to address our ordinary, everyday problems. Research is simply a quest to make our daily living easier as WE traverse our socio-economic society with solutions to ordinary day-to-day problems.

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Research seeks to explore an aim with specific objective(s) that must:

- a. Pose a domain question
- b. Review previous works about the problem space,
- c. Describe the adopted cum adapted method/technique/heuristic to be used vis-à-vis from the perspective of the dataset,
- d. Provide algorithm (step-by-step mode from problem to solution) on how achieving these specific objective(s) leads to the aim,
- e. Report findings/result,
- f. Explain findings as decision support framework provisioned from previous studies, system constraints and conditions under which the research was carried out – so that it can/may be

repeated

g. Provide recommendations and trade-offs (of adopted/adapted method) as implemented

If we belong to the R-&D unit – and tasked with developing strategic plans to market a recent product our company has made so that it competes favorably alongside similar products by so competitor companies. Some of the questions we can ask includes (not limited to these):

- a. What is the best strategy to promote the sales of this particular product?
- b. How many sales-persons do I need?
- c. What is the effect of a particular advertising campaign on the sales of this product?
- d. How satisfied are the consumers of this product?
- e. How much are consumers prepared to spend on this item?
- f. What do consumers like or dislike about this product?
- g. What type of packaging do consumers prefer for this product?
- h. What training do the sales-persons need to promote the sale of this product?
- i. What are the attributes of a good sales-persons?

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Characteristics of Research

From definitions, it is clear that research is a process of collecting, analysing, and interpreting data to answer questions. So, for any quest to qualify as a research, the processes must have certain characteristics such as:

- a. Controlled – Even though we can have a 1-to-1 relation – our daily living is quite complex and even more so are the relationships we build. Many factors play critical parts to yield a specific outcome. To study the nature of outcome, we must explore cause-and-effect relations, and be able to link effect(s) to its cause(s), and vice-versa. Thus, in the quest for control, we must set up our study to only explore the causality as restricted to two variables in a way that minimizes the effects of other factors affecting the relations. So, we must carefully observe how a method or its effects (i.e. dependent variable) impacts on a certain cause, domain task or problem (i.e. independent variable). This is quite true for scientific processes since they are often conducted in laboratories. However, because it is quite difficult to accomplish and control the external factors in social science – one may attempt to quantify their impacts.
- b. Rigorous – you must carefully and scrupulously ensure that procedures followed to find answers to the questions are relevant, appropriate for the task, and justifiable.
- c. Systematic – The procedures used and adapted to undertake the investigation follows a certain logical sequence. The different steps (within the algorithm) should not be taken haphazardly. It must meet with the criteria for the technical procedure and follow it to the later.
- d. Empirical – It implies that any conclusions drawn are based on hard evidence gathered from data that is collected from real life experiences and observations.
- e. Valid/Verifiable – It implies that whatever conclusions are reached (based on empiricism) are correct and can be verified through the repeat of such experiment. Thus, in your study – you

must describe a step-by-step means to repeat the process of arriving at the conclusion you did.

f. Critical – It is a detailed, proper scrutiny of the methods, tools and procedures used in your study. The process of investigation must be foolproof and free of any drawbacks. The process adopted and adapted to the study – are procedures that must withstand any critical scrutiny in the course of time.

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Types of Research

Research is usually classified into three (3) perspectives

- a. The application of the research study
- b. The objectives in undertaking the research
- c. The type of data sought

These are not mutually-exclusive. It implies a research classified from the viewpoint of application, can also be classified from the perspectives of 'objectives' and 'type of data sought'. These can also be done vice-versa

7/20/2023 Figure 1.1. The Types of Research 5

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Application: By this – there are two (2) broad categories:

- a. Pure research – involves the developing and testing of theories and hypotheses that are intellectually

challenging to the researcher; But, may (not) have any practical application at the present time or in the

future. Thus, it involves developing, examination, verification, and refinement of research methods, procedures, techniques and tools that forms a body of research methodology.

- b. Applied research – involve data collection about a domain problem, and using research techniques,

procedures and methods that form the body of research methodology to describe the nature of the data

collected in another way that often represents policies formulation, and enhancement of understanding

of a phenomenon.

Objectives – are broadly classified into descriptive, correlational, explanatory and exploratory

- a. Descriptive attempts to systematically describe or provide info about a phenomenon, an event, service or program. E.g. the needs of a community, the impacts of blockchain on a tracer system, attitudinal effects of phishing on undergraduates in Nigeria.

- b. Correlational attempts to discover and establish the existence of a relations, association and

interdependence between 2 or more aspects of a situation. E.g. Impact of personality traits in Phishing

attacks, what is the relationship between technology and employment? In all, these studies examines if

there are relations between two or more aspects of a situation.

c. Explanatory seeks to clarify why and how there is a relationship between 2 or more aspects of a situation or phenomenon. It tries to explain why technology as of today impacts employment of personnel in a company, explains how personality traits make users susceptible to phishing attacks.

d. Exploratory seeks to investigate the possibilities of undertaking a particular research. It is often referred to as a 'feasibility or pilot study'. It is usually carried out when a researcher wants to explore

areas about which s(he) has little or no knowledge about. A small-scale is undertaken to decide the worthiness of a more detailed investigation. On the basis of the assessment made during this exploratory study, a full study may eventuate. Thus, exploratory studies are used to develop, refine and

test measurement tools and procedures.

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Type of Data Sought

We have qualitative or quantitative – with three (3) criteria for the classification as thus:

□ The purpose of study – it is classified as qualitative if the purpose is primarily to describe the event, phenomenon or problem. The data is gathered through the use of variables measured on nominal or ordinal scales (qualitative measurement scales); and if the analysis is done to establish the variations in the situation, phenomenon or event without quantifying it. The description of an observed event, the historical enumeration of a phenomenon, an account of different opinion etc – are all qualitative research.

But, if we quantify the variation in an event with the data gathered using predominantly quantitative variables and if the analysis is geared to ascertain the magnitude of the variation – the study is classified as quantitative. E.g. Using Deep learning neural network to detect distributed denial of service attacks – seeks to classify (by quantifying) attacks from normal data packets.

□ Measuring the Variables – Use of statistics is not an integral part of quantitative or qualitative study designs. The function thus, of statistics is to act as a test to confirm or contradict the conclusions drawn on the basis of your understanding of analyzed data. Statistics, among other methods, helps us to quantify the magnitude of an association or relation, provide indication of confidence you can place in your findings and helps you to isolate the effects of different variables.

□ Analyzing the Data: Both qualitative and quantitative analysis have their strengths and weaknesses with none more superior. The measurement and analysis of variables about which

info is obtained – are dependent on the purpose of the study. In many studies, we may have to combine both qualitative and quantitative design approaches. E.g. to uncover the service available to undergraduate victims of phishing attacks and the extent of their vulnerability. The type of services available is a qualitative measure and aspect; while, the extent of the vulnerability using personality traits, media presence and others – are all quantitative aspects of the research design.

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The Essence of Research

The essence of reporting a research must be to:

- a. Encourage the repetition of such a research
- b. Provide new insight and new ideas for reference purpose and documentation
- c. Provide a new language that educates us in a new direction for decision support
- d. Encourage proper criticism (where possible)
- e. Advance the knowledge frontiers and domain

Knowledge discovery is at the heart of research. It is an interactive and iterative process that showcases the following characteristics:

- a. A sound philosophy that uses deductive reasoning processes
- b. Research is based on subjective insights reached via sound and verifiable results.
- c. Research requires an inter-disciplinary approach
- d. Spawn from a desire to be/do better. It is subjective, intangible and not an exact science.
- e. Research is based on interdependencies of causes-and-effects
- f. It is not a mechanical process. Researches are often rippled with known and unknown variables. It often can be described as a Problem P that requires a solution or goal G that is reached from a set of states S – allowing us to assign the values E to a corresponding set of variables V using known techniques, tools and methods.

The Aim of Research

Every research aims to advance knowledge about a domain. It achieves this by exploring techniques embedded within tools, and exploits historic data to provide us with new insights into new ideas that we can reference and document. This is because – research often involves detailed observations, theorizing with or drawing up hypotheses, experimenting to test the theorems or hypotheses, drawing up conclusions based on evidence presented, and reporting the findings – via a (social) scientific method.

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Knowledge Discovery In Research Data

Figure 1.2. Knowledge Discovery in Data

Figure 1.3. The Research Lifecycle

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Thus, research must aim to be (Specific-Achievable-Measurable-Reusable-Time Bound)

□ Specific - lists what a research aims to deliver so that WE know when it has reached its goal(a). It must address: who gains from the study? What is the expected/desired change? In what area of the society will the desired change impact? When will it happen? For example – This study is expected to cause a 60% reduction in production cost – making it more affordable for low-income earners in Delta State.

□ Measurable helps us to evaluate the stage(s) in reaching the goal. WE use benchmarks to measure success. E.g. Our performance during Exams, are measured and classed as A, B, C, D, E or F. So, a 60% reduction means that the researcher will prove s(he) has found a way to make the cost cheaper.

□ Attainable – answers the question of whether a research objective is attainable or can be achieved based on the resources available before the researcher. Thus, an objective must be coined/crafted to objectively access the resources available and accessible to us during the time of the research.

□ Relevant – tells us how important the work is and what consequences or impact the research portends to have on both the immediate society and the world at large.

□ Time-Bound: Timing is a critical factor in every research. An open-ended work or research may never be completed. Thus, WE must set up realizable aim/goal and (objectives) that must be accomplished within a certain time frame. Objectives are milestones, which when we achieve – leads us closer to our goal.

The Function of any/every research is to discover/contribute to Knowledge – even if such addition is just a single-step away. It changes from “what was” to “what is” now.

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Objectives of Outcome-Based Research

Research has three (3) main objectives namely:

- a. Theoretical Objectives are researches that seek to formulate new theories, principles, or laws. They are often explanatory in nature as it explains the relationships of certain variables. They also contribute to basic, human knowledge. E.g. includes researches in i.e., Physics, Computing, Chemistry, Mathematics, etc – all have theoretical objective(s).
- b. Factual Objectives seek factual finding, and the resolution of new facts. This objective is by nature descriptive. They describe previous facts (events) in history, etc.
- c. Application Objectives often do not contribute to new knowledge; Rather, they suggest

new applications, improvement and modification in practice. For example, if anyone gives a new application of electricity then such type of research has an application objective.

Types of Outcome-Based Research

Research types are based on various philosophies as thus in Table 1:

Scientific/Positivist Non-Scientific/Interpretivist

1. Action 1. Subjective/Argumentative
2. Laboratory/Field Experiments 2. Reviews
3. Surveys 3. Action Research
4. Case Studies 4. Case Studies
5. Theorem Proofs 5. Descriptive/Interpretive
6. Prediction/Classification/Regression 6. Future Research
7. Simulation 7. Role/Game Play Theorems

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Paradigms of Research

There are 2-main paradigms (positivist and naturalistic). Positivist – is rooted in physical sciences and lends itself to values that are systematic and scientific. Conversely, the naturalistic paradigm lends itself to qualitative, ethnographic and ecological approach. However, be it quantitative and/or qualitative design approaches - the authors using any of the paradigms – must be able to concisely make a distinction by adhering to values regarding the control of bias and the maintenance of objectivity in lieu of the research processes and the conclusions drawn. It is the application of these values to the process of data gathering, its analysis to yield results, discussion of these results and interpretation to draw conclusions – that enables it to be called a research process.

Outcome-Based Research Methodology

This is a set of systematic technique or guide used in conducting as well as reporting research that we have carried out. It aims to: (a) describe and analyse methods for performing research, (b) what actions can (and not) be done, (c) reveal the research limitations as a way-forward, (d) inform us of the needed resources for the kind of research, (e) clarify any assumptions and consequences our research may wish to make, and (f) relatively explore how their potentials and benefits can be harnessed to become twilight zones for cutting-edge and frontier knowledge.

Some of the many benefits of research methodology includes:

- Advances wealth of knowledge
- Provides tools, processes and techniques to perform research
- Develops critical thinking alongside with scientific attitude to observations – as the study outcomes scientific deductions and inductive reasoning with evidence
- Enriches the research processes as well as provides greater opportunity for insightful studies that will finally yield a better understanding of the domain subject
- Helps to inculcate the ability to evaluate and use research results with reasonable confidence and in decision making

□ Inculcate the ability to learn to read and think critically

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The Philosophy of Research Methodology

This is a belief about the way we critically, think through the research – geared towards solving the problem at hand. It also involves the way data is collected for the domain task, how it is analyzed, and used to generate the resultant findings. It also refers to the techniques adapted, the rationale for our desired choice and what relevant short/long-term contributions are advanced by the research. Philosophy can also depict the way research is conducted, and conceived, the strategy used and the research instrument used in pursuit of the research aim/goal, and the solution to the problem (research question).

There are basically two (2) types of research philosophies namely:

- Positivist beliefs adheres and describe the fact that reality is stable as observed from an objective viewpoint. It posits an event or phenomenon that is studied without interference. The conflict therein, is that observations made on such event, should be repeatable. And this often involves the manipulation of reality using independent variables. It is the basis of scientific methods. It is applied and associated with the physical and natural sciences.
- Interpretivist beliefs, contends that reality can be fully understood only via the subjective interpretation of and intervention. In addition, this belief also contends that the study of the phenomenon in their natural environment – is the key and also acknowledges scientists cannot avoid affecting the phenomena being studied.

The various philosophies have the various types of research as in Table 1.

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The Research Design Processes

This refers to how we intend to modes used in handling the research design processes:

Quantitative Research Design Qualitative Research Design

Scientific (Traditional): Qualitative (Traditional):

- Problem Statement □ Identify phenomenon and context of interest
- Observation and Experiment □ Theoretical basis and methodology
- Inductive Generalization □ Identify samples for research
- Hypotheses □ Data Collection/Gathering
- Attempted Verification of Hypotheses □ Data Analysis and Theory Formulation
- Proof or DisProof □ Knowledge in Context
- Knowledge Contributed

Scientific (Popper):

- Problem Statement
- Deduction of Testable Proportions
- Tests: Attempted Refutations
- Preference established between competing theories
- Knowledge Contributed

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The Pre-Dispositions of Research Methodology

This refers to the adopted and/or adapted research design used for the task at hand.

Quantitative Research Method/Design Qualitative Research Method/Design

Research Assumptions Made: Research Assumptions Made:

- Facts are presented as an objective reality □ Reality is socially constructed to show the relationship between
- The method to be used becomes the focal point of the interacting variables research (i.e. primacy of the method) □ Primacy is focused on the subject matter (i.e. variables) and their
- Variables can be identified and thus, the relationship relations rather than on the adopted method can be measured □ Variables are complex and interwoven; Thus, they are difficult to
- Results are measured in line with benchmarks. Thus, measure the outsider's point of view is quite critical □ Results express the researchers' point of view

Research Purpose: Research Purpose:

- Purpose of research can be generalized □ The purpose of the research is to contextualize
- Most research is geared toward the prediction □ Research is geared toward Interpretation
- Results can be causally explained □ The result/outcome is usually the researcher's perspective

Research Approach: Research Approach:

- Begins with hypotheses and theories □ Ends with hypotheses and grounded theories
- Research can be manipulated and its variables □ Results emerging from the analyses and variables are portrays a controlled semblance of the modeled system.
- The research uses formal instruments □ Research posits the views of previous/other researchers as
- It is mainly an experimentation design and is instrument cum basis for takeoff deductive by nature □ It is mainly naturalistic and deductive by nature
- The research uses component analysis □ It searches for patterns in its analysis
- The research seeks a consensus as to its norm □ The research seeks a consensus as to its norm
- Reduces data to numerical indices □ Seeks pluralism and complexity in its study
- Abstract languages can be used in report □ Makes minor use of numerical indices
- Uses descriptive languages in the report

Research Investigator's Role: Researcher Investigator's Role:

- The researcher must be detached and impartial □ Researcher show personal involvement and bias cum partiality
- His result must portray objectivity of analysis □ Result must be empathetic and understanding of the views sought.

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The active roles of a research investigator includes

- He should have a full understanding of the functions and activities of his job.
- He should have reflective thinking about various dimensions of his job activities.
- He should be sensitive, creative and imaginative to formulate action hypotheses for his problem
- He should have insights and teaching experience to identify the real problems at hand
- Must have a scientific attitude toward studying/observing things and be objective in his thinking
- He should have knowledge and skill in measuring instruments and elementary statistics.
- He should have an open mind so that he can discuss his problems with his colleagues and experts in the field to have a correct picture of the problem.
- He should have an urge to bring about excellence in job economical performance.
- He should be economical in designing the project from a time, energy, and money point of view.

Steps In Conducting an Action Research

The various steps in conducting research involves

1. Identify and Itemize the problems
2. Define the Aim of the Research and its specific objectives
3. Analyse the cause of the problems
4. Formulate the hypotheses based on the causes of the problems
5. Design the Method for Data Collection and Analysis
6. Find the Results and Discuss them in relation to the hypotheses for the problem as stated
7. Conclude by accepting/rejecting the hypotheses – and prescribing recommendations

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Figure 1.4: The Research Process

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Research method is taught as a supporting subject in many citadels using several means in many academic disciplines and at various levels by people committed to a variety of research paradigms.

- Step 1: Formulate a Research Problem: A research problem identifies your destination. It should tell any reader (your mentor and others) what you intend to research. It must be clear

and specific, and it guide how you must follow the research process such as study design, measurement procedure, sampling strategy, mode of analysis and style of writing your report (i.e. project, dissertation or thesis). It is important to evaluate your research problem in light of the financial resources at your disposal, time available, your disposition and your supervisor/mentor's expertise cum knowledge in the field of study. It is equally important to identify the gaps in your knowledge of relevant discipline such as statistics as required for the analysis. Thus, ask yourself if you have sufficient enough knowledge about computers and programming in the domain problem (if you wish to use them) – see Lecture Note 2.

□ Step 2: Conceptualize Research Design: Research involves scientific methods that are systematic, controlled, valid and rigorous establishment of causes-and-effects association that permits accurate prediction of outcomes under a given set of conditions. It also involves verifications of what is already known, identification of past errors and limitations of previous studies, identifying the gaps in knowledge as well as identifying techniques and technologies to be used in resolving the gaps/errors. The validity of what you find rests solely on how it is found. The main function of a research design is to explain how you will find the answers to your research questions. The research design sets out the logic of your inquiry; And thus, should include: (a) the study design, (b) logical arrangement of how you propose to undertake it, (c) the measurement procedures, (d) sampling strategy, (e) the mode of analysis, and (f) duration for the analysis. For any investigation, the selection of an appropriate design is crucial to arriving at a valid finding, comparisons and conclusions. Faulty design leads to misleading result and waste of time and financial resources. The strength of an empirical investigation is evaluated on the design which must be valid, workable and manageable.

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□ Step 3: Construct Instrument for Data Collection: A research instrument or tool is what we use in collecting data. E.g. observation form, questionnaire, interview guides etc. Creating the appropriate tool for data collection is a first practical guide to carrying out your study. Data can either be primary (data you collect yourself primarily for the purpose of the study), or secondary (already collected and stored data). So, be it primary or secondary data, you must develop an instrument (to extract the required data for your study) and field test the instrument on a similar sample data. Do not field test the instrument on your sample study. If you are also using computers for data analysis – you must provide the means and mode for coding/translating the data on to it computed form. This helps us to determine: (a) required data, (b) noisy data, and (c) partial truth data that may introduce uncertainties and ambiguities into your study.

□ Step 4: Sample Selection: Result accuracy depends on the sample data selected and selected features in dataset. The goal of sampling is to minimize (within cost limits) the gaps between the values obtained from your samples and those prevalent in population. Sampling ensures that even if relatively small unit is scientifically selected, it provides with sufficiently high probability, a true representation of the population under study. Sampling seeks to avoid bias

and attain maximum precision for a given outlay of resources. The 3 types of sampling are: random, non-random/probability sampling, and mixed sampling.

□ Step 5: Writing a Report: Having collected the data and analyzed it – WE then must put everything together so that it provides adequate information to its reader. The overall plan is to tell a reader: (a) about the problem, and (b) how you plan to investigate the problem. The first task is a research proposal – whose function is to detail the operational plan for obtaining answers to your research problem. Thus, a proposal ensures and reassures the reader of the validity of the methodology to obtain answers that are accurate and objective. Thus, a proposal must contain: (a) what you propose to do, (b) how you plan to proceed, and (c) why you selected the proposed strategy. Thus, the proposal should contain these information:

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Thus, the proposal should contain these information:

- A background of the problem that helps the reader understand what you wish to do
- Study's problems and limitations as identified
- Aim of the study, and specific objectives (attained) will lead to the aim/goal of the study
- A list of hypothesis, if you are testing any
- A review of previous works, what you know and the errors in previous works
- A list of clearly and concisely identify gaps in the previous knowledge
- A study design you are proposing to use and setting for your study
- The research instrument you plan to use, data on sample size and sampling design
- An outline on data processing procedures
- Financial resources / budgetary plans
- References

□ Step 6: Collecting Data: We have a problem, study design, research instrument and selected sample

data; Next, we must collect the data to be used to make inferences and conclusions. You must decide

how to collect the data and which features within the dataset collected is to be used.

□ Step 7: Processing Data: How we analyse the data depends largely on: (a) the type of data (i.e. descriptive, quantitative, qualitative, attitudinal), and (b) the way you want to write the report. So, you

must consider if it is quantitative/qualitative report; consider if the data is to be analyzed; how and what

type of analysis required (i.e. statistics procedures). Then, you must also identify the variables to be

subjected to these statistical procedures.

□ Step 8: Writing a Report: Your report informs the world of what you have done, what you have discovered and what conclusions you have drawn from your findings. Thus, making the step the hardest to accomplish. But, your mentor is meant to act as a support. So, listen always and follow

his/her guide. Your report should be written using an academic format/style with different chapters and sections based on themes. This differs for many citadels. Adhere to their preferred structure format.

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□ Planning a Research: A research must be reported – even if its expected outcome differs from what was proposed. Research results, positive or negative should be reported. They may provide insight to why such a result was obtained. Every research stems from a local issue. If we seek to provide a solution to a non-existent problem and/or issue – then YOU have simply, created one.

□ Choosing a Topic and the rest – This is quite the easiest and also the hardest process...

1. Topic must sustain you interest
2. Must be manageable
3. Avoid topics that others could not complete
4. Pick from courses you loved
5. Poses it as a Question
6. Time frame matters for every topic
7. Google and download (pdf and doc formatted) materials
8. Create a Grammarly Account to use to proofread your research
9. Create a TurnItIn or iThenticate Account through your university or a friend
10. Install Mendeley, Zotero, EndNote, X9 for in-text citations and referencing
11. Write and Present 3-topic using the Proposal Format

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Moving Forward – is really up to you. No one forced you here. But, if you must go ahead with it – then come with us. Starting is always the easy part – the hard part is staying on-course. A major component of this initial planning to stay on course is that you (as the researcher) must create a ToDo List of the following:

- a. Break down research into many Chapters for the Dissertation / Thesis
- b. Count Weeks to submission deadline. Create Gantt Chart guide to completing the Research. Ensure

you are always 2-weeks ahead of the chart – as everything will move faster than you can recall.

- c. Work out these weeks to be spent on each chapter bearing in mind these:
 - i. Google and download materials relating to the topic – nothing less than 50 articles
 - ii. Research these materials, write and submit to the mentor/supervisor
 - iii. Every paragraph from Chapters 1 – 4, should be supported with in-text references or citations to

which author(s) said what – as you have done nothing substantial in the field to become a quoted authority. Do not ever quote yourself.

iv. Supervisor/Mentor to read through your submission(s) and revert back to you

v. Always effect mentor's suggestions. Never miss/ignore mentor's suggestion(s). Ask question(s) if you do not understand the suggestions.

vi. Repeat the process till the mentor/supervisor is satisfied

vii. Check-off the completed (mentor-satisfied) chapter

viii. Proceed with the next chapter.

e. Design/modify your slides for your viva-voce. Time is not on your side

f. During presentation – take notes and ask questions on suggestions. Don't argue with anyone at all.

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Meeting your Mentor

A mentor is a beacon, pointer, or focal point. Someone you reach out to guide and validate some opinions you have developed or reached. S(he) becomes a feedback to your research.

Knowing his/her research area can help strengthen yours. Your mentor is your guide and cannot effectively lead you if his/her interest differs from what you are researching on. So, find a research passion in which you can be properly guided. Your research must often become a mix between "what you love" and your mentor's interest. Ensure your mentor fixes a time for you and be there always.

1. Present your 3-topics proposal

2. Mentors are resources for your use

3. Make notes of their suggestion

4. Book appointments & be there (no excuses)

5. Check if s(he) can Whatsapp/Zoom. Do not assume s(he) must oblige you such excesses.

6. Avoid your Mentor's Dos & Dents.

7. Ask Questions, if confused

8. Discuss your ideas/suggest path

9. Be serious always with this work

10. Always meet a Mentor's Deadline

11. No matter how good you are, your Mentor has something to offer to the research and the research is

much of his/her work, just as it is yours. But, more importantly – s(he) will present you as ready for viva

12. Alongside Proposal, show your Proposed Structure

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Outcome-based Research and Reporting

Lecture 03: Step 2 - Conceptualize a Research Design

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The Research Design

This is a plan, structure and strategy of investigation so conceived as to obtain answers to a research question. The plan is the complete scheme or program of research. It usually includes an outline of what an investigator is to do from writing the hypotheses and their operational implications

– to the final analysis of the data. It is a procedural plan that a researcher adopts to answer questions validly, objectively, accurately and economically. It is also a blueprint or detailed plan for how research study is to be completed – operationalizing variables so they can be measured, selecting a sample of interest to study, collecting the data to be used as a basis for testing the hypotheses, and analysing the results.

Functions of Research Design

It has 2-major functions namely: (a) to conceptualize an operational plan to undertake the various procedures and tasks required to complete a study, and (b) ensure the procedures are adequate to obtain valid, objective and accurate answers to the research question.

A research design thus should:

1. Name a study design that is cross-sectional, before-and-after, comparative, control experiment, or random control, and

2. Provide detailed info about the following:

☐ Who will constitute the study population?

☐ How will the study population be identified?

☐ Will a sample or the whole population be selected?

☐ If a sample is selected, how will it be contacted?

☐ How will consent be sought?

☐ What method of data collection will be used and why?

☐ In case of questionnaires, where will responses be returned to? How should respondents contact you if they have queries?

☐ In case of interview, where will it be conducted?

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Commonly Used Study Designs

The various designs have been classified by examining them from 3-perspectives as thus

- Number of contacts with the study population
- The reference period for the study
- The nature of the investigation

The Number of Contacts – based on this feat, designs have been classified into 3 namely:

□ Cross-sectional is best suited for studies aimed at finding prevalence of an event. They help us see an overall picture of the study as it stands at time of the study. It is simple in design and helps you decide what you want to find, identify the study population, select a sample, and contact the respondents to provide the required data. They involve only one contact with the study population – they are comparatively cheaper to undertake and easy to analyse. They cannot measure change because we need 2-observations at 2-points in time on the same population.

□ Before-and-After is best suited to measure the impact or effectiveness of a system. Thus, it describes 2-sets of cross-sectional observations on the same population to find out the change in the situation between 2-points in time. Change is measured by comparing the differences in the state(s) at an before/after the observations. It demerit is that (a) 2-sets of data is collected, (b) time lapse may result in respondent attrition, (c) reactive effect as instrument can sometimes educate the respondents, and (d) maturity effects can change the study population from one phase to another.

□ Longitudinal studies are best suited for studying the pattern of change in relation to time. Thus, here – the sample population is visited a number of times at regular intervals. It has the same demerits of the before-and-after studies.

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The Reference Period – is the time frame in which a study is exploring a phenomenon or event. These are often categorized from three (3) perspectives namely:

- Retrospective studies investigate events of the past. They are conducted either based on the data available for that period, or due to the respondents that recall the situation. E.g the use of the ABACUS computer system in Pre-historic America
- Prospective studies refer to likely prevalence of an event in the future. Such studies seek to establish the outcome of an event or what is to likely happen. Experiments are usually classified as prospective studies since the researchers must wait for an intervention to register its effects on the study population. E.g. of topics can be: (a) to determine, under field conditions – the impact of XGBoost on the classification of contents using sentiment

analysis, (b) empirical evidence to classify credit-card transactions using modular deep learning neural network etc.

□ Retrospective-prospective studies focuses on past trends in an event, studies it into the future and then drawn conclusions from its findings as cause and relations of the past to simulate its impact now and forecast the future. A study is classified thus when you measure the impact of an intervention without having a control group. Most before-and-after studies, if carried out without having a control group where the baseline is constructed from the same population before introducing the intervention – are classified to be in this category. Thus, it implies a part of the data is collected retrospectively from the existing records before the intervention is introduced and then the study population, is followed to ascertain the impact of the intervention. Trend studies, which becomes the basis for projections – falls into this category. E.g. effects of random breath testing on road accidents, impact of incentives on the productivity of employees of an organization, effects of advertisements on sales of a product.

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The Nature of the Investigation – Studies by nature of the investigation are classified into 3. To understand the difference, we consider some examples:

□ Suppose, we wish to test the impact of the a(synchronous) teaching/learning style on the level of comprehension of students. In any situation, there is first – the cause-and-effect relationship. There are 2-ways to studying this relationship.

First way, involves a researcher introducing the intervention, which is the cause of the change. As with our example, the 2-modes of learning: (a) synchronous (face-2-face) mode where a teacher passes on knowledge to various learners at the same time, and (b) asynchronous mode involves the use of media-assisted learning where learners can learn anything, anywhere and anytime. Thus, the researcher introduces the learning style and waits till it has produced a certain effects – or it is given a sufficient amount of time to produce – change. Second way, a researcher observe an event and attempts to establish what caused the change. Thus, s(he) starts with an outcome and seeks to establish what caused it.

Thus, for the nature of investigation, we can describe study designs into experimental, non-experimental and quasi/semi experimental as thus: Any study that uses the first mode (i.e. starts with a cause to establish the effects) is classified as experimental. Note that in this mode, the independent variables can either be observed, introduced, controlled or manipulated by the researcher. Conversely, where a study starts with an effect and traces the cause, it is grouped as non-experimental – and the independent variables cannot be manipulated as the assumed cause has already happened/occurred. Instead, the researcher seeks to retrospectively link the cause(s) to the outcome(s). A semi/quasi experiment design has the features of both the experimental and non-experimental design

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Figure 3.1. Experimental and Non-Experimental Studies Design

The Experimental Study Design

We consider some experimental studies design as thus:

- The after-only design
- The before-and-after design
- The control design
- The double-control design
- The comparative design
- The placebo design

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□ After-only design: A researcher knows the data population will be, or is already exposed to an intervention and wishes to study its impact. The data baseline (pre-test/before observation) is constructed on the basis that respondents recall the event before the intervention, or from data available in records. Change in the dependent variable is measured as a difference between before (baseline) and after observations. Technically, it is faulty design to measure the impact of intervention as there is no baseline data to compare an after observations with. Its demerit is that: (a) the 2-sets of data are not comparable, (b) design measures total change, including the change due to extraneous variables, (c) it cannot identify the total effect of an intervention. But, this design mode is widely used in impact assessment studies, because some programs operate without the benefit of a planned evaluation. Its adequacy depends on having reasonably, accurate data about the prevalence of an event before the intervention was introduced. E.g impact of random breath testing on road accidents, impact of advertisements on the sales of a product, the impact of the declining mortality rate on the fertility of a population etc. Here, it is expected that accurate data is kept about the event under study, and any change is primarily due to the advent of the intervention or change in the policy.

7/20/2023 Figure 3.2. The after-only design 7

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□ The before-after design: It overcomes the issue of retrospectively constructing the before observation by establishing it before the intervention is introduced to the study population. Then, when the service has been implemented, or is assumed to have had its effect – the after observation is carried out to ascertain the impact due to the intervention. This design takes care of only one problem of the after-only design – that is, it provides the ground and basis for

comparability of the before and after observations. It still does not enable a researcher to conclude that any change (in whole or a part) can be attributed to the program intervention. To overcome this, a control group is used. The before-and-after design may also from the problems identified in the after-only design. The impact of the intervention in the before and after design is calculated as thus:

Change in dependent variable = [status of dependent variable at the after observation] - [status of the dependent variable at the before observation]

7/20/2023 Figure 3.3. The before-and-after design measurement of change 8

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□ The control-group design: For such study, the researcher selects 2-population groups (as against one).

One is termed as control and the other is termed experimental. These groups are expected to be comparable as far as possible in every respect except the intervention. The experimental group either

receives or is exposed to the intervention – whereas, the control group is not. First, before observations

are made on both groups at the same time. The experimental group is then exposed to the intervention.

When it is assumed that the intervention has had its impact, an after observation is made on both groups.

Any difference in the before and after observations between the groups regarding the dependent variable(s) is then attributed to the intervention. For the experimental group, the total change in the

dependent variable is given by:

$$(Y_e) = (Y''_e - Y'_e)$$

Where Y''_e = after observation on experimental group, and Y'_e = before observation on experimental group

$$(Y_c) = (Y''_c - Y'_c)$$

Where Y''_c = post-test observation on control group, and Y'_c = pre-test observation on control group.

The total change is given by: Impact of intervention = $(Y''_e - Y'_e) - (Y''_c - Y'_c)$. Please, note that the chief

objective is to quantify the impact of extraneous variables and ascertain the impact of the intervention only.

7/20/2023 Figure 3.4. The control-group experimental design 9

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□ The double-control design: Although the control design helps the researcher to quantify the impact that can be attributed to extraneous variables – it does not separate out other effects that

may be due to research instrument (such as reactive effects), or respondents (such as maturation or regression effects). When the researcher needs to identify and separate out these effects, a double-control design is required. Thus, in a double-control study, the researcher has 2-control groups instead of one. To quantify – say, the reactive effects of an instrument, the researcher excludes one of the control group from the before observation. The difference is computed thus:

$$(Y_e) = (Y''_e - Y'_e)$$

Y''_e = after observation on experimental group, and Y'_e = before observation on experimental group

$$(Y_{c1}) = (Y''_{c1} - Y'_{c1})$$

Y''_{c1} = [Impact of extraneous variable on control group 1] $\pm$ [reactive effects] $\pm$ [Random effects]

$$(Y_{c2}) = (Y''_{c2} - Y'_{c2})$$

Y''_{c2} = [Impact of extraneous variable on control group 2] $\pm$ [reactive effects] $\pm$ [Random effects]

Thus, Reactive Effect = $(Y''_{c1} - Y'_{c1}) - (Y''_{c2} - Y'_{c2})$.

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Figure 3.5. The double-control-group experimental design

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□ Comparative design is best if a researcher wish to compare the effectiveness of different services modes. It allows a study to either be experiment or non-experiment. In comparative experimental design, the study population is divided into the same number of groups as the number of treatments/services to be tested. For each group, the baseline for dependent variable is established. The different treatment models are then introduced to the various groups. After a certain period, when it is assumed that the treatment models have had its effect, the after observation is performed to ascertain any change in the dependent variable. The degree of change in the different population is then compared to establish the relative effectiveness of the various intervention.

But, for a non-experimental comparative design, groups already receiving different interventions are identified, and only the post-observation is conducted. The pre-test observation is constructed either by asking the population in each group to recall the required information relating to period prior the introduction of the treatment, or it is extracted from existing records. Sometimes, pre-test observation is not constructed at all on the assumption that if the group are comparable, the baseline must be identical. As each group is assumed to have the same baseline, the difference in the post-test observation is assumed to be – because of the intervention.

7/20/2023 Figure 3.6. The comparative design 11

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□ Placebo design attempts to determine the extent of this effect – it involves 2 or 3 groups, depending on whether or not the researcher wants to have a control group. If and when the researchers decides to have a control group, the first group receives treatment, the second

group receives the placebo treatment and the third group (which is the control group) receives nothing. The decision as to which group will be the treatment, the placebo or control group, can also be made through randomization.

Figure 3.7. The Placebo Design

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Questions

- ☐ Describe the nature of a research design and state its 2 major functions.
- ☐ Based on number of contact features, describe the 3-types of study design you know.
- ☐ Based on reference period, describe the types of study design you know
- ☐ Based on the nature of the study design, describe the types of study design you know.
- ☐ Using the experimental design type, describe the workings of any of these
 - a. After-only experimental design
 - b. The before-and-after design
 - c. The control design
 - d. The double-control design
 - e. The comparative design
 - f. The matched control experimental design
 - g. The placebo design
- ☐ State other known designs that can be used.

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